

Exponents and Scientific Notation

41 points

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Laws of Exponents9 points

1. Write each expression as a single power.

a. $4^3 \cdot 4^5$ 1 pt

b. $(2^3)^4$ 1 pt

c. $\frac{7^9}{7^4}$ 1 pt

d. 9^0 (give its value) 1 pt

2. Find the value of each expression.

a. $\left(\frac{2}{3}\right)^3$

1 pt

b. $6^0 \cdot 5^2$

1 pt

c. $\frac{(10^2)^2}{10^3}$

1 pt

3. Which expression is equivalent to $2^4 \cdot 2^3$?

1 pt

- ☐ A. 4^7
- ☐ B. 2^{12}
- ☐ C. 4^{12}
- ☐ D. 2^7

4. Find the value of n that makes the statement true: $\frac{5^n}{5^2} = 125$

1 pt

2. Integer Exponents

11 points

5. Rewrite each expression with positive exponents. Do not evaluate.

a. 6^{-3} 1 pt

b. $\frac{1}{(-2)^{-5}}$ 1 pt

c. $3 \cdot 10^{-4}$ 1 pt

6. Find the value of each expression.

a. 4^{-2} 1 pt

b. $(-3)^{-3}$ 1 pt

c. $\left(\frac{1}{2}\right)^{-3}$ 1 pt

d. $2^5 \cdot 2^{-7}$ 1 pt

7. Which is the value of 5^{-2} ?

1 pt

- ☐ A. $\frac{1}{10}$
- ☐ B. -25
- ☐ C. $\frac{1}{25}$
- ☐ D. -10

8. Select ALL of the expressions whose value is $\frac{1}{64}$.

1 pt

- ☐ A. 2^{-6}
- ☐ B. 4^{-3}
- ☐ C. -8^2
- ☐ D. 2^{-8}
- ☐ E. $\left(\frac{1}{4}\right)^3$

9. Marco says the value of 3^{-2} is -9 . Explain why he is wrong, and give the correct value. 1 pt

10. Which is greater, 2^{-3} or 3^{-2} ? Show how you know.

1 pt

3. Variable Bases

9 points

11. Write each expression as a single monomial.

a. $x^4 \cdot x^6$ 1 pt

b. $(y^3)^5$ 1 pt

c. $(3a^2)^3$ 1 pt

d. $\frac{15m^7}{5m^2}$ 1 pt

12. Rewrite each expression using only positive exponents.

a. b^{-4} 1 pt

b. $5x^{-2}$ 1 pt

c. $(2m^{-3}n)^2$ 1 pt

13. Which expression is equivalent to $(4x^3)^2$?

1 pt

- ☐ A. $16x^5$
- ☐ B. $8x^6$
- ☐ C. $4x^6$
- ☐ D. $16x^6$

14. How are x^2 and x^{-2} related? Explain, and give the value of each when $x=5$.

1 pt

4. Scientific Notation

12 points

15. Write each number in scientific notation.

a. 8,300,000

1 pt

b. 0.00047

1 pt

c. 60,500,000,000

1 pt

d. 0.0000000092

1 pt

16. Write each number in standard form.

a. 2.9×10^5 1 pt

b. 7.1×10^{-6} 1 pt

17. Which is 0.0036 written in scientific notation? 1 pt

- ☐ A. 3.6×10^3
- ☐ B. 3.6×10^{-3}
- ☐ C. 3.6×10^{-2}
- ☐ D. 36×10^{-4}

18. Answer each question.

a. How many times larger is 9×10^9 than 3×10^6 ? 1 pt

b. How many times larger is 4.8×10^{-3} than 1.6×10^{-7} ? 1 pt

19. Which is greater, 8.9×10^{-4} or 2.1×10^{-3} ? Explain how you know. 1 pt

20. A computer displays a number as 5.6E8. Write the number in standard form.

1 pt

21. Between which two consecutive powers of 10 does 4.7×10^{-5} lie?

1 pt